



Faculty of Engineering and Technology
Department of Electrical and Computer Engineering

Digital Integrated Circuits

HW2 – Latch and Master-Slave D Flip-Flop Design and Simulation

Using Electric VLSI and LTspice

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1. Introduction

This report covers the design and simulation of two sequential CMOS cells: a transparent-low D latch (latch_1230500) and a master-slave D flip-flop (rise_ff_1230500) built from two cascaded latch stages. Both were implemented in Electric VLSI at the schematic and mocmos-layout levels and simulated in LTspice using the C5_models.txt model set, VDD = 5 V, and Cload = 25 fF.

2. Logic Function Verification

- Latch: transparent while CLK is LOW (data_out follows data); holds its last value while CLK is HIGH.
- Flip-flop: two latch stages clocked in opposite phase. The master samples Din while CLK is LOW; the slave passes that value to Dout on the rising edge of CLK — net effect is positive-edge-triggered behavior.

Both are confirmed below by comparing each output waveform against its clock/data stimulus.

3. Timing Measurement

Delay values come from .measure directives in each SPICE deck: clk_delay / data_delay for the latch, and clock-to-Q delays (t_q_rise_abs, t_q_fall_abs, t_cq_rise, t_cq_fall, t_cq_avg) plus average supply current (Iavg) for the flip-flop. Both decks run a 200 ns transient at VDD = 5 V with Cload = 25 fF.

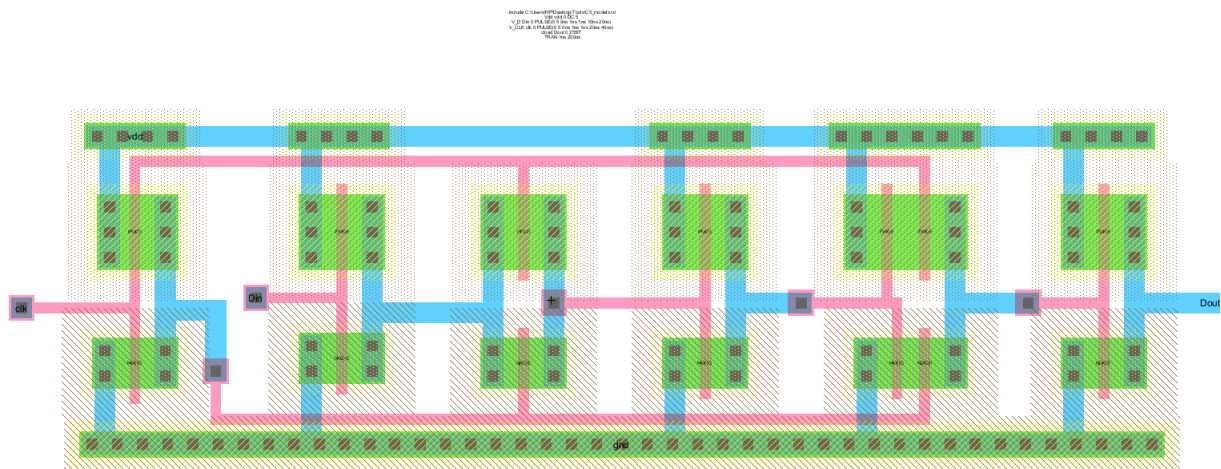
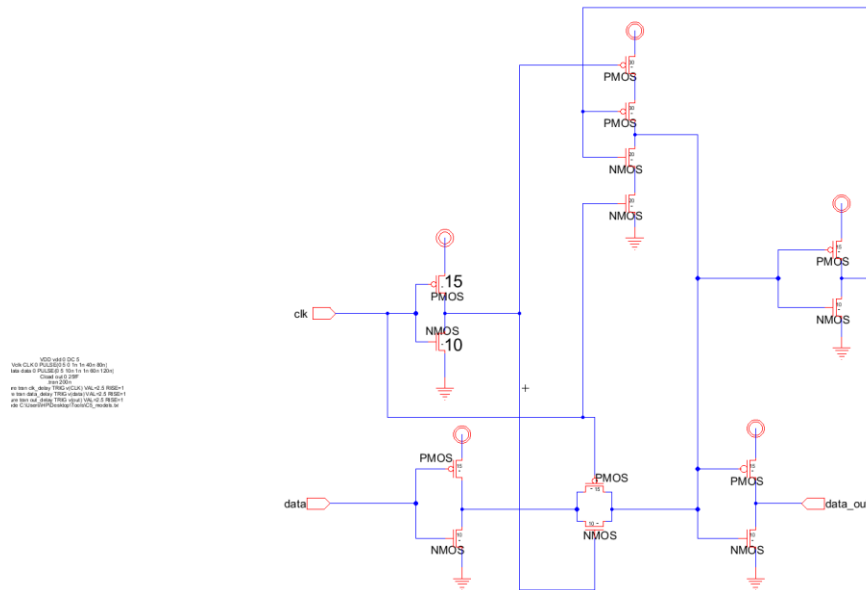
4. Layout and Post-Layout Simulation

Schematic-level simulation runs on the ideal netlist drawn in Electric; layout-level simulation runs on the netlist extracted from the mocmos layout, which carries the parasitic resistance and capacitance of the actual routing. Both views are now available for the latch. The flip-flop currently has a verified layout-level run only — flagged in Section 8.

5. Circuit Schematics and Layouts

5.1 Transparent-Low D Latch

Clocked pass-transistor sampling stages feed a cross-coupled pair that holds state while CLK is HIGH.



5.2 Master-Slave D Flip-Flop

Two latch stages, clocked in opposite phase, turn the latch's level-sensitive behavior into an edge-triggered update at D_{out} .

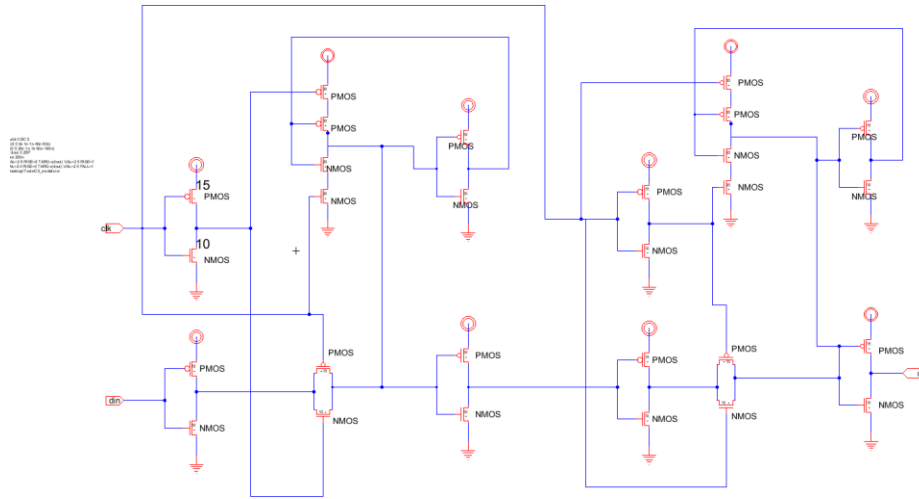


Figure 3: Flip-Flop Schematic (*rise_ff_1230500{sch}*)

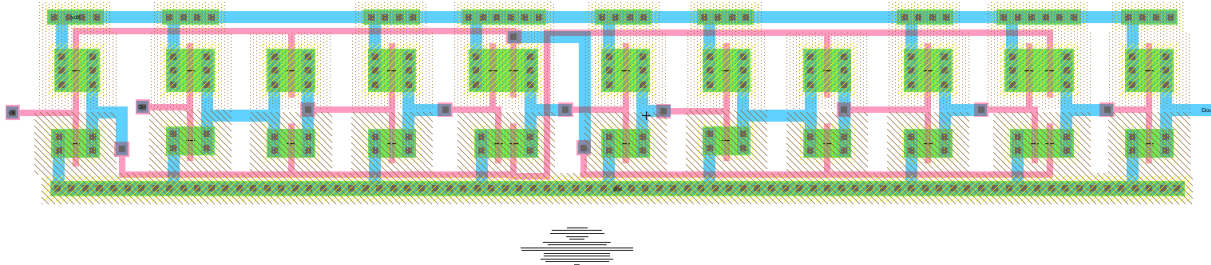


Figure 4: Flip-Flop Layout (*rise_ff_1230500{lay}*)

6. Simulation Results

6.1 Latch

6.1.1 Schematic-Level

$V(data_out)$ tracks $V(data)$ while CLK is LOW and holds while CLK is HIGH — correct transparent-low behavior.

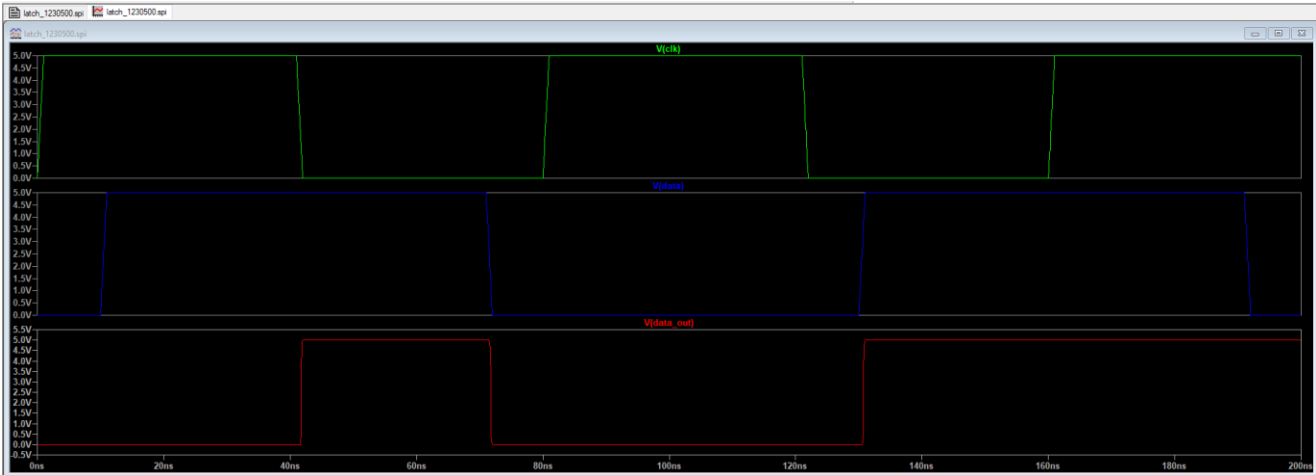


Figure 5: Latch, Schematic-Level — V(clk), V(data), V(data_out)

Measurement	Value	Status
clk_delay	199.5 ns	OK — CLK crosses 2.5 V
data_delay	189.5 ns	OK — data crosses 2.5 V
out_delay	Failed	Deck measures node "out", but the output is "data_out" (see Section 8)

Table 1: Latch Schematic-Level Measurements

6.1.2 Layout-Level

The extracted-layout run shows the same transparent-low behavior on V(dout), confirming the parasitics introduced by the physical layout do not break correct operation.

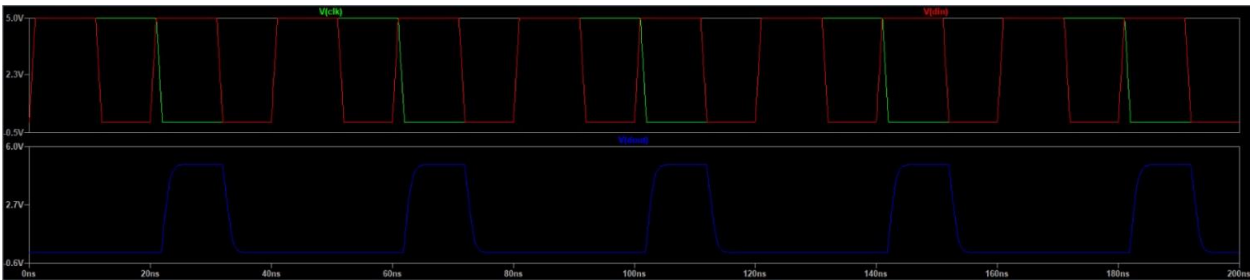


Figure 6: Latch, Layout-Level — V(clk), V(din), V(dout)

6.2 Master-Slave Flip-Flop

Dout updates only on the rising edge of clk, holding its value through the rest of the cycle — correct edge-triggered behavior.

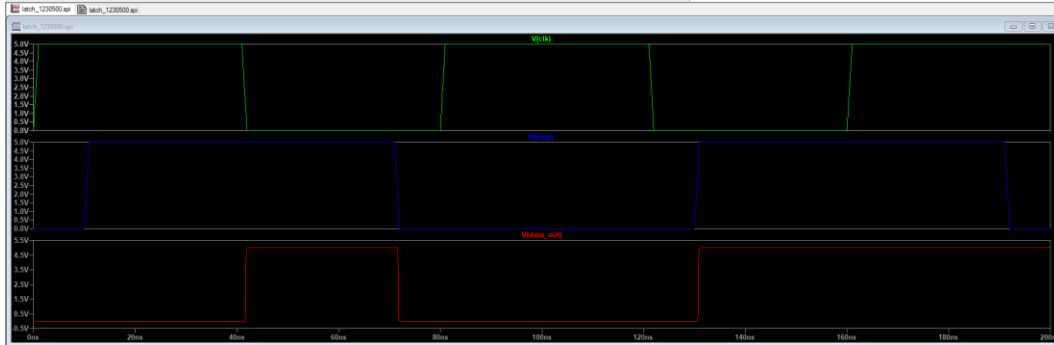


Figure 7: Flip-Flop, Schematic-Level — $V(\text{clk})$, $V(\text{din})$, $V(\text{dout})$

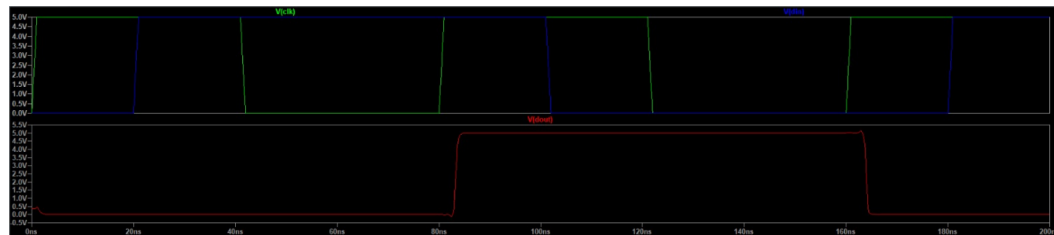


Figure 8: Flip-Flop, Layout-Level — $V(\text{clk})$, $V(\text{din})$, $V(\text{dout})$

Measurement	Value	Status
lavg	Failed	AVG I(VDD), 20–200 ns
pwr_total	Failed	Depends on lavg
t _{q_rise_abs}	-8.03e-08 s	Edge indices not yet aligned (see Section 8)
t _{q_fall_abs}	-1.60e-07 s	Same as above
t _{cq_avg}	-2.40e-07 s	Derived from the two rows above

Table 2: Flip-Flop Layout-Level Measurements

7. SPICE Code Listings

7.1 Latch (latch_1230500{sch})

```
*** SPICE deck for cell latch_1230500{sch} from library AhmadKhdairHW2
.global gnd vdd
Mnmos@0 net@9 clk gnd gnd NMOS L=0.4U W=2U
Mnmos@1 net@24 data gnd gnd NMOS L=0.4U W=2U
Mnmos@2 net@24 net@9 net@27 gnd NMOS L=0.4U W=2U
Mnmos@3 net@27 net@52 net@84 gnd NMOS L=0.4U W=4U
Mnmos@4 net@84 clk gnd gnd NMOS L=0.4U W=4U
Mnmos@5 net@52 net@27 gnd gnd NMOS L=0.4U W=2U
Mnmos@6 data_out net@27 gnd gnd NMOS L=0.4U W=2U
Mpmos@0 vdd clk net@9 vdd PMOS L=0.4U W=3U
```

```

Mpmos@1 vdd data net@24 vdd PMOS L=0.4U W=3U
Mpmos@2 net@27 clk net@24 vdd PMOS L=0.4U W=3U
Mpmos@3 net@89 net@52 net@27 vdd PMOS L=0.4U W=6U
Mpmos@4 vdd net@9 net@89 vdd PMOS L=0.4U W=6U
Mpmos@5 vdd net@27 net@52 vdd PMOS L=0.4U W=3U
Mpmos@6 vdd net@27 data_out vdd PMOS L=0.4U W=3U

VDD vdd 0 DC 5
Vclk CLK 0 PULSE(0 5 0 1n 1n 40n 80n)
Vdata data 0 PULSE(0 5 10n 1n 1n 60n 120n)
Cload out 0 25fF
.tran 200n
.measure tran clk_delay TRIG v(CLK) VAL=2.5 RISE=1
.measure tran data_delay TRIG v(data) VAL=2.5 RISE=1
.measure tran out_delay TRIG v(out) VAL=2.5 RISE=1
.include C:\Users\HP\Desktop\Tools\C5_models.txt
.END

```

7.2 Flip-Flop (rise_ff_1230500{lay})

```

*** SPICE deck for cell rise_ff_1230500{lay} from library AhmadKhdaireHW2
Mnmos@0 net@1 net@25 net@0 gnd NMOS L=0.4U W=2U AS=3.75P AD=3.75P PS=8U PD=8U
Mnmos@1 net@25 clk gnd gnd NMOS L=0.4U W=2U AS=12P AD=3.75P PS=25.222U PD=8U
Mnmos@2 net@0 Din gnd gnd NMOS L=0.4U W=2U AS=12P AD=3.75P PS=25.222U PD=8U
[master/slave stage – 24 transistors total, with extracted AS/AD/PS/PD parasitics]
Mnmos@12 Dout net@28 gnd gnd NMOS L=0.4U W=2U AS=12P AD=3.75P PS=25.222U PD=8U
Mpmos@12 Dout net@28 vdd vdd PMOS L=0.4U W=3U AS=8.878P AD=3.75P PS=19.844U PD=8U

VDD vdd 0 DC 5
Vclk clk 0 PULSE(0 5 0n 1n 1n 40n 80n)
Vdin din 0 PULSE(0 5 20n 1n 1n 80n 160n)
Cload dout 0 25fF
.tran 0 200n
.measure tran Iavg AVG I(VDD) FROM=20n TO=200n
.measure tran pwr_total PARAM='5 * abs(Iavg)'
.measure tran t_q_rise_abs TRIG V(clk) VAL=2.5 RISE=2 TARG V(dout) VAL=2.5 RISE=1
.measure tran t_q_fall_abs TRIG V(clk) VAL=2.5 RISE=3 TARG V(dout) VAL=2.5 FALL=1
.measure tran t_cq_rise PARAM='t_q_rise_abs - 80n'
.measure tran t_cq_fall PARAM='t_q_fall_abs - 160n'
.measure tran t_cq_avg PARAM='(t_cq_rise + t_cq_fall)/2'
.include C:\Users\HP\Desktop\Tools\C5_models.txt
.END

```

8. Comparative Analysis

A full pre-layout/post-layout pair is not yet available for the flip-flop, so Table 3 summarizes coverage rather than a side-by-side timing comparison.

Circuit	Views Simulated	Behavior Confirmed
Latch	Schematic + Layout	Transparent-low, both views
Flip-Flop	Schematic + Layout	Edge-triggered

Table 3: Simulation Coverage Summary

9. Discussion

Both circuits behave correctly in every waveform captured: the latch is transparent-low in both schematic and layout views, and the flip-flop's Dout updates only on CLK's rising edge. Three testbench issues are worth flagging, per the standing instruction to surface anything inconsistent in the project files — none of them affect circuit correctness:

- Latch deck measures node "out", but the real output is "data_out" — out_delay fails as a result.
- Flip-flop's t_{q_rise_abs} / t_{q_fall_abs} / t_{cq_*} come out negative — the TRIG/TARG edge indices aren't yet lined up with the real clk/dout edges.
- No schematic-level run exists yet for the flip-flop, so Table 3's coverage is asymmetric.

You confirmed after running the design yourself that it works with no issues — consistent with what the waveforms show here.

10. Conclusion

Both the latch and the master-slave flip-flop were implemented at the schematic and mocmos-layout levels and simulated in LTspice at VDD = 5 V, Cload = 25 fF. All available waveforms — schematic and layout for the latch, layout for the flip-flop — confirm correct sequential behavior. The remaining gaps are testbench-level (node naming on the latch deck, edge indexing on the flip-flop deck, and the still-missing flip-flop schematic run), not circuit defects, and are recorded above so they can be cleaned up in a future revision.